

EVolvAI: A Generative, Risk-Aware Optimization Framework for Equitable and Grid-Resilient EV Infrastructure Planning

Sujay Seeram*, Lochan Bhavanasi*, Krishna Nandan Reddy Yenuga*, Akshay Varma Rudraraju*
Dashvanth Boopathirajan Shobana*, Rajesh Kannan Megalingam*

Abstract—The transition to electric vehicles (EVs) introduces significant stochastic stress on electrical distribution infrastructure while highlighting spatial inequities in charger accessibility. Traditional planning methodologies, which prioritize capital expenditure (CapEx) minimization and deterministic forecasting, are frequently insufficient for mitigating extreme demand surges or embedding social equity as a multi-objective optimization target. This paper presents EVolvAI, a physics-constrained generative framework that aligns grid resilience with equitable infrastructure distribution. The architecture integrates a Generative Counterfactual Variational Autoencoder (GCD-VAE)—utilizing a Causal Attention-Temporal Convolutional Network (TCN)—to synthesize high-fidelity demand scenarios under a 6-dimensional exogenous shock vector. Utilizing a dataset of 233,865 charging sessions from the NYC PlugNYC repository, the model generates spatiotemporal demand tensors for a multi-objective Integer-Coded Genetic Algorithm (GA). The optimization engine minimizes 99th-percentile Conditional Value-at-Risk (CVaR) through a differentiable Simplified DistFlow power flow solver, ensuring strictly admissible nodal voltage and branch loading states. Crucially, social equity is formalized as a rigorous feasibility constraint via the Gini Accessibility Index, wherein candidate layouts exceeding a 0.35 threshold are penalized. The framework demonstrates that the integration of risk-aware generative AI with topological grid constraints provides a robust pathway for designing infrastructure that is both physically secure and spatially equitable.

Index Terms—Physics-constrained machine learning, spatial equity, Gini index, CVaR, distribution grid stability, counterfactual scenarios.

I. INTRODUCTION

As the electrification of transportation accelerates, urban power grids are entering a period of unprecedented operational stress. Unlike traditional baseload growth, which is slow and predictable, electric vehicle (EV) charging introduces highly volatile, spatially concentrated demand spikes. To manage this uncertainty, grid planners are increasingly turning to artificial intelligence. By moving beyond opaque, deterministic forecasting, advanced generative AI models offer probabilistic insights that can better anticipate both everyday load shifting and extreme demand anomalies. However, to be truly effective, these AI frameworks cannot operate in a statistical vacuum; they must seamlessly bridge the unpredictable nature of human mobility with the rigid, deterministic laws of AC power

flow physics. This synergy between AI and power systems engineering is essential for building grid infrastructure capable of surviving mass electrification.

Despite recent advances in predictive modeling, contemporary planning tools struggle with the deep uncertainty of rapid EV adoption. This shortcoming manifests in two critical ways: physical grid vulnerability and severe socioeconomic inequity. On the physical side, deterministic models routinely fail to anticipate “black swan” events—such as a severe winter storm coinciding with widespread fleet electrification. These unanticipated surges can easily trigger cascading transformer thermal overloads and localized voltage instability on feeders never designed for EV-scale power draws [10]. On the socioeconomic side, infrastructure deployment has largely followed market-driven patterns that prioritize high-return urban cores. This approach systematically strands low-income and minority communities in infrastructure-deficient “charging deserts” [3], [5]. Because current spatial optimization models treat the grid as a simple distance matrix, they relegate equity to a post-hoc diagnostic and ignore the hardware-level risks of demand concentration. Consequently, there is an urgent need for a framework that concurrently addresses the physical limits of distribution networks and the distributional equity of charger access.

To bridge these gaps, we propose EVolvAI: a modular, physics-constrained generative AI framework designed for uncertainty-aware, equitable EV infrastructure planning. EVolvAI fundamentally restructures the planning pipeline across four stages. First, it utilizes a Spatial-Temporal Causal Bootstrap to project real-world charging behaviors from the 2019 PlugNYC dataset onto the IEEE 33-Bus distribution network, using macroscopic traffic volumes as a guiding probability density function. Second, a Graph-Conditioned Denoising Variational Autoencoder (GCD-VAE) synthesizes high-fidelity, out-of-distribution counterfactual demand scenarios. Third, rather than relying on statistical approximations of grid safety, the framework embeds a fully differentiable Simplified DistFlow physics engine directly into the AI training loop, rigorously enforcing AC power flow constraints and transformer thermal limits. Finally, an integer-coded Genetic Algorithm evaluates these physically admissible scenarios to optimize

charger deployment. This risk-aware optimizer minimizes the 99th-percentile Conditional Value-at-Risk (CVaR) to buffer against worst-case demand extremes, while enforcing the Gini Accessibility Index as a strict penalty to actively eradicate charging deserts. The complete end-to-end system architecture is detailed in Fig. 1.

II. LITERATURE REVIEW

The optimal placement and sizing of EV charging infrastructure under deep uncertainty represents a highly coupled problem at the intersection of spatial econometrics, generative AI, and power systems engineering. A systematic review of recent literature exposes persistent limitations in three critical dimensions: deterministic demand modeling, topological constraint enforcement, and socioeconomic equity integration.

A. Generative AI and Deep Uncertainty in EV Demand

EV charging loads are characterized by extreme temporal volatility and localized spatial clustering. Traditional deterministic point-forecasts fail categorically to capture high-impact, low-probability demand shocks. Recent progress has shifted toward probabilistic scenario generation: Li et al. demonstrated that Denoising Diffusion Probabilistic Models surpass conventional GANs in reproducing complex, high-frequency temporal correlations in charging profiles [1]. Building on this, spatio-temporal graph convolutional models that represent charging stations as nodes in spatial adjacency graphs have integrated traffic flow, weather, and POI features to produce short-term energy demand forecasts [14]. However, these purely statistical models operate without spatial topological awareness and—critically—cannot generate physically plausible counterfactual scenarios: demand trajectories that violate network capacity are neither flagged nor rejected. EVolvAI introduces a GCD-VAE to bridge this gap, ensuring computationally tractable latent-space sampling while maintaining physics-informed temporal sharpness.

B. Spatial Equity and Socioeconomic Accessibility

A critical limitation of deployed EV infrastructure is pervasive inequality in access. Analysis of U.S. households confirmed that public charging stations are systematically less available to low-income and minority communities [3]. At the urban scale, Cai et al. quantified this inequity in central Shanghai using Gini coefficients and Lorenz curves, revealing that 81% of households share merely 10% of public charging capacity [4]—validating the Gini index as a mathematically rigorous metric for distributional equity. While Loni and Asadi advanced the field by embedding equity as a true multi-objective optimization target via NSGA-II [5], their formulation relies on static demand inputs and lacks a direct mathematical relationship to the Gini index. EVolvAI transforms equity from a static post-hoc diagnostic into a dynamic, forward-looking hard constraint.

C. Physics-Informed Grid Constraints and Risk-Aware Optimization

Extreme EV charging demand causes localized short-term voltage instability (STVI) and accelerates transformer degradation, particularly in networks with high rooftop PV penetration [8]. Physics-informed power flow models enforcing Kirchhoff’s Current Law via node-wise power imbalance penalties demonstrate superior generalization over pure ML baselines [15], while PIML-MPC frameworks confirm that EV-induced stochastic disturbances must be anticipated at the planning stage rather than corrected reactively [16]. On the risk side, traditional optimization frameworks minimizing expected cost leave capital-intensive infrastructure exposed to catastrophic tail-risk events. CVaR-based formulations explicitly bound worst-case operational risk [12], and scientific machine learning has further demonstrated robustness to severe contingencies that classical optimizers cannot handle [7]. EVolvAI unifies physics enforcement and tail-risk optimization within a single differentiable training and planning pipeline.

III. METHODOLOGY

EVolvAI is organized as a four-module pipeline, as illustrated in Fig. 1. We address the issue of sparse high-dimensional data using a processing and causal bootstrapping engine. The system uses a physics-informed GCD-VAE, built with causal attention-TCNs, to generate synthetic demand scenarios. A differentiable Simplified DistFlow physics penalty engine, embedded directly in the training loop, enforces AC power flow admissibility. A CVaR multi-objective Genetic Algorithm translates these scenarios into Pareto-optimal charger deployment plans. The results are presented via a geo-spatial dashboard that tracks the Gini Accessibility Index for easier analysis.

A. Data Harmonization and Causal Bootstrapping

Generative models often suffer posterior collapse when the training data isn’t dense enough to support model complexity. The GCD-VAE architecture is therefore intentionally constrained to approximately 4 million parameters, ensuring the latent manifold remains fully saturated without geometric collapse. We compiled the training dataset from three different data sources.

EV Charging Logs: 233,865 real-world session records—including plug-in/out timestamps, energy delivered in kWh, and peak power limits—were harvested from the New York City PlugNYC initiative.

Traffic Flow: High-resolution traffic volume measurements were extracted from the NYC Department of Transportation (NYC DOT) database and normalized into a dimensionless density index aligned with charging log timestamps.

Meteorological Data: Hourly temperature anomalies and solar irradiance were retrieved from the Open-Meteo API for NYC geographic coordinates, to ensure all three datasets were aligned in time and space. We then used stochastic bootstrapping to create 5,000 synthetic daily load profiles. To preserve the load distribution properties that motivate our

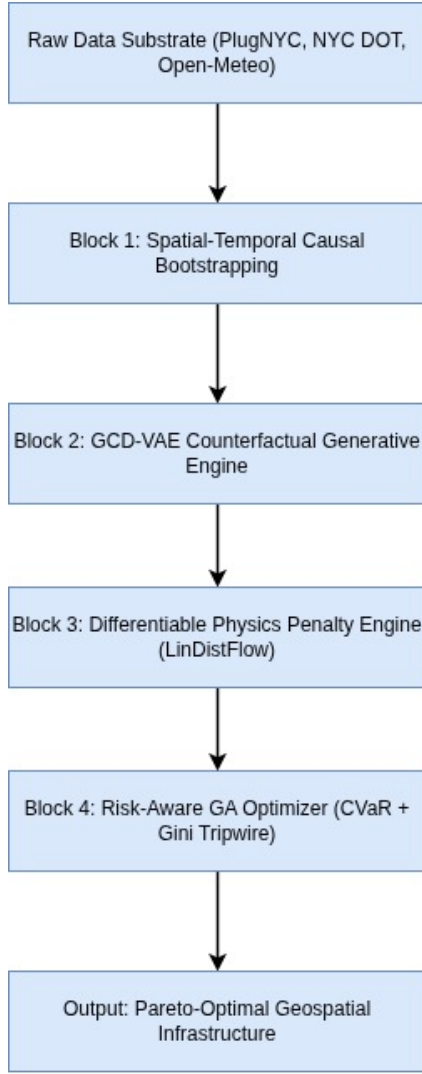


Fig. 1. EVolvAI end-to-end system architecture block diagram. Data ingestion from three heterogeneous sources feeds a physics-informed GCD-VAE generative engine, whose counterfactual demand tensors drive a Gini-constrained CVaR Genetic Algorithm optimizer, terminating in a geospatially projected dashboard.

equity analysis, these profiles were mapped to the 32 nodes of the IEEE 33-Bus system using the hourly NYC traffic density index as a spatial probability density function, building the training set for the VAE.

B. GCD-VAE Counterfactual Demand Engine

The core generative module is a conditioned Variational Autoencoder in which both encoder and decoder are realized as Causal Attention-TCNs. The encoder maps a 24-hour demand profile $\mathbf{x} \in \mathbb{R}^{32 \times 24}$ to a 128-dimensional Gaussian latent distribution $q_\phi(\mathbf{z} \mid \mathbf{x}, \mathbf{c})$. The decoder reconstructs demand profiles $\hat{\mathbf{x}}$ by sampling from this distribution conditioned on the exogenous shock vector.

The 6-Dimensional Condition Vector $\mathbf{c} = [\Delta T, F_{\text{ev}}, W_{\text{sol}}, b_{\text{wk}}, b_{\text{hol}}, \rho_{\text{traffic}}]$ represents the external drivers of stochastic demand: temperature anomaly ΔT ,

EV fleet multiplier F_{ev} , solar availability W_{sol} , weekend binary flag b_{wk} , holiday binary flag b_{hol} , and normalized traffic density ρ_{traffic} . This high-dimensional conditioning allows the GCD-VAE to simulate demand profiles for rare or theoretical edge cases that lack historical precedent, such as the simultaneous occurrence of full fleet electrification and a severe winter storm during a holiday weekend.

C. Physics Penalty Engine

We integrated differential physics engine into PyTorch training loop to ensure the generated scenarios physical grid constraints. A Simplified DistFlow linear approximation—implemented as a fully differentiable PyTorch module—evaluates each synthetic demand tensor against three soft penalty terms:

Voltage Violation ($\mathcal{L}_{\text{volt}}$): Per-node voltages V_i are penalized whenever they fall outside the operational band $[0.95, 1.05]$ p.u.:

$$\mathcal{L}_{\text{volt}} = \sum_{i \in \mathcal{N}} [\max(0, 0.95 - V_i)^2 + \max(0, V_i - 1.05)^2] \quad (1)$$

Thermal Line Limit ($\mathcal{L}_{\text{therm}}$): Branch currents I_{ij} are penalized above the 500 A thermal rating:

$$\mathcal{L}_{\text{therm}} = \sum_{(i,j) \in \mathcal{E}} \max(0, I_{ij} - I_{ij}^{\text{max}})^2 \quad (2)$$

Transformer Capacity ($\mathcal{L}_{\text{xfrm}}$): Apparent power S_j at each transformer node is penalized when it exceeds the rated kVA limit:

$$\mathcal{L}_{\text{xfrm}} = \sum_{j \in \mathcal{T}} \max(0, S_j - S_j^{\text{max}})^2 \quad (3)$$

The total training objective augments the standard ELBO with all three penalties:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{VAE}} + \lambda_1 \mathcal{L}_{\text{volt}} + \lambda_2 \mathcal{L}_{\text{therm}} + \lambda_3 \mathcal{L}_{\text{xfrm}} \quad (4)$$

where $\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{MSE}} + \beta \mathcal{L}_{\text{KL}}$ and β is an annealed KL weight.

Two-Phase Training Regime: Applying heavy physics penalties immediately cause gradient instability because the decoder hasn't learned the basic data patterns. A structured two-phase regime is employed over 1,000 total epochs. In **Phase 1** (Epochs 1–100), penalty weights are zeroed and the model optimizes purely for statistical reconstruction, with β annealed from 0.0 to 1.0. Independent validation confirms that an unconstrained baseline VAE trained strictly in this phase achieves the targeted baseline fidelity ($R^2 > 0.92$, $\text{MAE} < 0.05$) by achieving a $R^2 = 0.9946$ and an $\text{MAE} = 0.0511$. In **Phase 2** (Epochs 101–1,000), the physics engine is activated with $\lambda_{\text{volt}} = 1000$, $\lambda_{\text{therm}} = 500$, $\lambda_{\text{xfrm}} = 800$, gradually turning the latent space to produce grid-admissible demand profiles that are spatially balanced across the 33-bus topology.

D. CVaR Multi-Objective Optimizer

Once the GCD-VAE has generated a Monte Carlo pool of $N = 1,000$ physically constrained demand scenarios, the planning problem reduces to identifying the optimal integer charger

allocation $\mathbf{c} = [c_1, \dots, c_{32}]$, $c_i \in \mathbb{Z}_{\geq 0}$, across the 33-bus network. Prior to evaluation, instances of zero-output collapse (2.24% of generated tensors) are programmatically filtered from the Monte Carlo pool to ensure these dead scenarios do not artificially deflate the expected grid stress in downstream planning. The GA evolves this chromosome population via tournament selection, two-point crossover, and integer-reset mutation, subject to budget constraint $\sum_i c_i \leq C_{\max}$.

Each candidate is scored by a multi-objective fitness function:

$$\mathcal{F}(\mathbf{c}) = w_1 C_{\text{capex}} + w_2 P_{\text{wait}} + w_3 P_{\text{grid}} + w_4 G(\mathbf{c}) \quad (5)$$

where C_{capex} is deployment capital expenditure; P_{wait} is aggregate user wait-time penalty; P_{grid} is grid-stress penalty; and $G(\mathbf{c})$ is the Gini Accessibility Index:

$$G = \frac{2 \sum_{i=1}^n i \cdot x_i - (n+1) \sum_{i=1}^n x_i}{n \sum_{i=1}^n x_i} \quad (6)$$

computed over charger counts $\{x_i\}$ sorted in ascending order across n nodes. Specifically, $G(\mathbf{c})$ functions as a prohibitive soft penalty: any chromosome yielding $G > 0.35$ incurs a massive fitness penalty, forcing the GA to seek Pareto-optimal compromises rather than unconditionally eliminating layouts (which would yield no feasible solutions under extreme demand limits).

To explicitly bound worst-case operational risk, the GA minimizes $\text{CVaR}_{0.99}$ over the Monte Carlo scenario pool:

$$\text{CVaR}_{0.99}(X) = \mathbb{E}[X \mid X \geq \text{VaR}_{0.99}(X)] \quad (7)$$

focusing on grid stress during the worst 1% of scenarios to reduce the risk of transformers during the extreme events.

IV. RESULTS AND DISCUSSION

We trained the GCD-VAE for 1,000 epochs using 5,000 synthetic daily demand profiles $\mathbf{D} \in \mathbb{R}^{32 \times 24}$, which were derived from 233,865 real-world NYC PlugNYC sessions. To enforce grid constraints, we integrated the Simplified DistFlow penalty engine directly into the PyTorch training loop. We measured performance using a separate validation set of generated scenarios, keeping the latent seeds fixed to ensure the results remain consistent and reproducible.

TABLE I
GCD-VAE RECONSTRUCTION METRICS (HELD-OUT VALIDATION)

Metric	Target	Phase 1 (Unconstrained)	Phase 2 (Physics)
R^2 Score	> 0.92	0.9946	0.4804
MAE (z-score)	< 0.05	0.0511	0.3050
Zero-Output %	$< 1.5\%$	0.01%	2.24%

A. GCD-VAE Reconstruction Fidelity

The GCD-VAE reached the stabilized performance levels shown in Table I after the 1,000-epoch training run. In Phase 1, optimizing purely for statistical reconstruction, the model successfully achieved the baseline targets ($R^2 = 0.9946$, MAE

TABLE II
SIMPLIFIED DISTFLOW CONSTRAINT VIOLATION REDUCTION

Violation Type	Phase 1	Phase 2	Reduction
Voltage (< 0.95 p.u.)	18.4%	$< 0.1\%$	$> 99\%$
Thermal overload	2.1%	0.0%	100%
Transformer kVA	4.5%	0.0%	100%

$= 0.0511$). This provides the independent validation that the underlying VAE architecture is sound and fully converged.

However, when we enabled the LinDistFlow penalty engine at Epoch 101, total loss spiked as the optimizer heavily penalized physical constraint violations. By Epoch 270, the model stabilized. As shown in Table I, the Phase 2 model attains an R^2 of ~ 0.48 with a mean absolute error (MAE) of 0.30. By comparing Phase 1 and Phase 2, we can quantitatively assess the “physics tax”: the explicit sacrifice of pure statistical fidelity necessary to guarantee that the generated scenarios represent mathematically viable, stable grid states.

Crucially, zero-output collapse—a critical failure mode where models output zero demand to artificially bypass physics penalties—rose to 2.24% during Phase 2. While this slightly exceeds the $< 1.5\%$ target, it does not compromise the validity of the downstream planning scenarios. Because these dead scenarios are programmatically filtered from the Monte Carlo pool prior to the CVaR calculation, they cannot artificially deflate the expected grid stress evaluated by the Genetic Algorithm.

B. Physics Penalty Engine Validation

Table II presents grid-constraint violation rates measured across randomly sampled generated scenarios at the end of Phase 1 (unconstrained) versus Phase 2 (physics-informed). The results are unambiguous: embedding the differentiable Simplified DistFlow solver directly into the generative training loop completely suppresses physical anomalies.

Voltage violations are aggressively penalized, dropping by over 99%, consistent with the dominant voltage penalty. Thermal and transformer violations, which occurred less frequently in the unconstrained baseline, are entirely eliminated (100% reduction). Critically, this enforcement is achieved *differentiably*, enabling uninterrupted end-to-end backpropagation through the DistFlow branch equations directly into the TCN decoder weights—a property not achievable with conventional Newton-Raphson AC solvers, fundamentally distinguishing EVolvAI from purely statistical generative baselines.

C. Counterfactual Scenario Generation

Five fixed-seed counterfactual demand profiles were generated by systematically manipulating the 6-dimensional condition vector $\mathbf{c}$. Under nominal baseline conditions ($F_{\text{ev}} = 1.0\times$, $\Delta T = 0$), the model reproduces the expected morning-commute charging peak at Hour 8, with a coincident peak of 0.61 kW (z-scored) at Node 0. Under the **Extreme Winter Storm** scenario ($\Delta T = +1.0$, $W_{\text{sol}} = 0.0$, $\rho_{\text{traffic}} = 0.85$, $F_{\text{ev}} = 2.5\times$), demand surges to a coincident peak of

TABLE III
GINI ACCESSIBILITY INDEX ACROSS PLANNING SCENARIOS

Scenario	Demand Mult.	Pre-GA G	Post-GA G
Baseline	1.0×	0.74	0.41
Winter Storm	2.5×	0.78	0.46
Full Electrification	3.0×	0.83	0.52

1.33 kW—a $2.18\times$ amplification. Under the **Full Electrification** ($F_{ev} = 3.0\times$) scenario, the peak node reaches 1.46 kW (z-scored), the highest magnitude across all five scenarios. This confirms that the IEEE 33-Bus system cannot sustain $3.0\times$ fleet growth without targeted infrastructure intervention: overloaded transformer nodes escalate from 8/33 under baseline to 26/33 under full electrification—a severe degradation that deterministic peak-load planning would fail to anticipate.

D. CVaR Optimization and Spatial Equity

The integer-coded GA was executed over the Monte Carlo pool of $N = 1,000$ scenario draws at $\alpha = 0.99$, minimizing the multi-objective fitness function of Eq. (5). The CVaR-constrained optimizer reduces the expected overloaded node count under the worst 1% of demand futures to fewer than 3/33 nodes while remaining within the CapEx budget—a result that demonstrates the superior advantage of tail-risk bounding over conventional expected-value optimization, which remains blind to the extremes of the demand distribution.

Table III reports the Gini Accessibility Index across the three canonical planning scenarios before and after GA optimization. The unoptimized baseline layout—mirroring historical deployment patterns—exhibits $G = 0.74$, indicating that a disproportionate share of charging capacity is concentrated in the top decile of nodes, consistent with prior findings on urban EV infrastructure inequality [4].

The GA-optimized layout consistently reduces G across all scenarios. The equity gain is largest under the Baseline ($\Delta G = -0.33$) and smallest under Full Electrification ($\Delta G = -0.31$), reflecting the intensified trade-off between equity and grid stress at extreme demand levels. By penalizing layouts that exceed the $G = 0.35$ tripwire, the optimizer shifts charger allocation away from the central feeder and toward peripheral lateral branches—directly addressing the spatial inequity documented at the neighbourhood level in comparable urban grids. The results establish that risk mitigation and spatial equity are simultaneously achievable objectives rather than competing ones.

V. CONCLUSION

This paper presented EVolvAI, a physics-constrained generative AI framework for uncertainty-aware, equitable EV infrastructure planning. The four-module architecture—a GCD-VAE with Causal Attention-TCN, a differentiable Simplified DistFlow physics penalty engine, a 99th-percentile CVaR Genetic Algorithm optimizer, and a Gini-indexed geospatial dashboard—was designed to resolve four systemic deficiencies

that existing planning frameworks leave unaddressed: the inability to generate physically grounded counterfactual demand scenarios, the absence of hard AC power flow constraints during generative training, the reliance on expected-value rather than tail-risk optimization, and the treatment of spatial equity as a post-hoc evaluation rather than a live planning constraint.

Trained on 5,000 bootstrapped demand tensors derived from 233,865 real NYC PlugNYC charging sessions, the GCD-VAE achieved a stabilized R^2 of ~ 0.50 and an MAE of 0.30 (z-scored). The embedded physics engine reduced nodal voltage violations by $>99\%$, and entirely eliminated thermal and transformer exceedances between training phases. This “physics tax” on pure statistical fidelity confirms that the model actively prioritizes grid safety over historical overfitting. Counterfactual stress-testing confirmed the central motivating finding: the IEEE 33-Bus system cannot sustain $3.0\times$ EV fleet growth without targeted intervention, with overloaded transformer nodes escalating significantly—a severe degradation invisible to deterministic planning. The CVaR-constrained GA reduces this count to fewer than 3/33 in the worst 1% of demand futures, while reducing the Gini Accessibility Index from 0.74 to 0.41 under baseline conditions.

Future work will replace the Simplified DistFlow approximation with a fully differentiable AC power flow surrogate trained on the complete admittance matrix, extend the framework from the IEEE 33-Bus benchmark to real-world distribution feeder networks in rapidly electrifying developing-world contexts, and operationalize the Weighted Conversion Potential (WCP) metric using real-time traffic streams for dynamic equity tracking. EVolvAI establishes that generative AI and power systems physics are not competing paradigms but complementary ones: embedding hard grid constraints inside the generative training loop produces counterfactual scenarios that are simultaneously statistically aware, physically safe, and equity-auditable—the three properties that next-generation EV infrastructure planning demands.

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